Hexbound: Game Concept and Implementation Plan

Team members:

Крикота Давид ІМ-33
Войтків Владислав ІМ-33
Розумовський Даниїл ІМ-34
Лев Береза ІМ-32

1. Overview

Hexbound is a single-player, turn-based fantasy roguelite for the browser. The player crosses a hand-made hex map towards a boss, chooses which locations to visit on the way, and fights with a deck of cards built along the route.

The player is the Wayfarer, a lone traveller crossing three lands to slay the Ember dragon. The game has three levels, and a full run takes about 40 minutes.

Course	«Основи розроблення комп'ютерних ігор»
Team	4 members
Development time	Under 2 months
Platform	Web browser
Genre	Single-player roguelite with route planning and card combat

2. Design pillars

Four pillars decide which features go into the game: a feature that serves none of them is cut.

Pillar	**What it means**
The route is the main decision	Where you go decides what you fight, what you earn and how strong you are at the boss.
The map feeds the fight	Terrain you crossed and places you visited change the next fight, not just your numbers.
Few rules, real choices	Every screen (a fight turn, a reward, a shop, an event) offers at least two sensible options.
Growth you chose	Starting weak, the player shapes a deck with a clear direction.

3. Core gameplay

Core loop

A level is one loop repeated tile by tile until the boss falls:

1. Plan a route on the map.
2. Step onto the next tile (each step costs food).

3. Resolve the location: a fight, a chest, a village, a shrine or an event.
4. Grow: take a card, a blessing, gold or supplies.

The route decides which fights you take and in what shape you arrive; the fights pay for the next part of the route.

Map and routes

Each level is a hand-made 9×7 hex map with the start and the boss in opposite corners, about 9–10 steps apart. Every map offers three readable routes: **direct** (short, passes an elite), **safe** (longer, with a village) and **rich** (a detour through chests and shrines).

- **Slot maps.** Terrain, start, boss and villages are fixed; other locations are dealt at random into marked slots each run, so a level is never a memorised replay.
- **Visibility.** Location types are visible from the start; details (which monster, chest contents) appear within 2 tiles. The boss and its signature move are shown from the start.
- **Food.** Each step costs 1 food (2 on slow terrain). A level gives enough food for the direct route plus about 3 spare steps, so every detour is a trade.
- **Terrain hazards.** Each level adds one hazard that also affects the next fight, for example ice chills you (−1 energy on the first turn). Every hazard can be avoided by a longer route.

Locations

Location	What happens
Monster	Fight a normal enemy; win gold and pick 1 of 3 cards.
Elite	Harder fight on the short route; better rewards and a key.
Chest	No fight; gold plus a card, food or healing.
Locked chest	Opens only with a key from an elite; holds a rare card and a blessing.
Village	Buy food, rest, buy or remove cards, take a bounty.
Shrine	Pick 1 of 3 permanent blessings.
Mystery	A choice event with its cost and dice odds shown on the buttons.
Boss	Hardest fight; ends the level.

Resources

The player tracks three meters (health, food and gold) plus 3 lives for the whole run, and shapes a build made of the deck and blessings. Gold earned on a level pays for about two village purchases, so the player must choose between healing, food and new cards.

Cards and combat

The deck has about 20–24 card types in four families: weapons, spells, armour and items. The player starts with 10 cards and leans into one of three build directions: Blade (weapons), Arcane (spells) or Bulwark (armour). Rules on cards use only four keywords: Freeze, Chill, Exhaust and Draw.

Fights are short, turn-based battles against one enemy whose next move is always shown. Each turn the player draws 5 cards and has 3 energy to spend on attacks, block and effects. Target length: 2–4 turns for a normal fight, 6–10 for a boss.

Levels and progression

Level	Hazard	Boss	What it tests
1. Whisperwood	Swamp (slow)	Goblin chief	Basics; doubles as the tutorial
2. Frostpeak Pass	Ice (slow, chills)	Frost giant	Route and food planning
3. Ashen Citadel	Lava (damage)	Ember dragon	Whether the deck has a real direction

Deck, gold and blessings carry over between levels. On death the player spends a life to retry the level; with no lives left, the run ends. The end screen shows a score so runs can be compared.

4. Scope

The must-have list is what the team commits to deliver; everything else is added only after a full run is playable from start to finish.

Must-have

- Three levels with hand-made terrain, slot maps and three routes each.
- Movement, food, terrain hazards and their effect on fights.
- All location types, including elites with keys, the locked chest, villages with card removal and bounties, and 6–8 choice events.
- Combat with visible enemy intents, 5-card hands and the four keywords.
- About 20–24 cards, 9 normal enemies, 3 elites, 3 bosses, 10–12 blessings.
- 3 lives, a score screen and a tutorial first fight.
- All content stored as data files, so it can be balanced without code changes.

Nice-to-have: extra chest types, card upgrades, sound and music, a consistent art set, saving a run between sessions.

Out of scope: multiplayer, random terrain generation, multiple playable characters, a story beyond short level texts.

5. Team roles

Each of the four members owns one area of the project.

Role	Responsibilities
PM / Game designer	Introduces the game concepts and rules, manages the team, priorities and implementation plan; designs cards, events and blessings.
UI / Level designer	Implements all screens, the interface and visual feedback; builds the three hand-made maps and the art.
Programmer	Codes the game: architecture, map and movement, combat and enemies, locations, lives and score, release build.
Tester	Finalises the concepts from the game designer into testable rules; tests each feature, runs playtests and checks balance against the design targets.

6. Implementation plan

The game is built in six stages. Each stage ends with an exit check, and the team moves on only when it passes. Durations are given as a share of the whole timeline.

Stage 0. Kick-off and decisions (~5%)

- **PM / Game designer:** presents the design, closes open questions (engine, art style), splits the must-have list into tasks.
- **UI / Level designer:** picks the visual direction and the list of screens; sketches the hex grid.
- **Programmer:** sets up the repository on the prototype, project structure and data-file format.
- **Tester:** reviews the design for gaps; turns the pillars into a playtest checklist.

Exit check: decisions made, task board ready, project builds and runs.

Stage 1. Playable first level (~25%)

- **PM / Game designer:** writes level-1 content (starter deck, enemies, boss) and the tutorial fight.
- **UI / Level designer:** map and fight screens; the Whisperwood map with three routes.
- **Programmer:** map, movement and food; combat core with intents, energy, 5-card hands and keywords.
- **Tester:** tests each system as it is added; first check of route choices.

Exit check: a teammate who did not build it can finish level 1.

Stage 2. Full game loop and content (~30%)

- **PM / Game designer:** final lists of cards, enemies, blessings, events and bounties; prices.
- **UI / Level designer:** village, shop, shrine, event, reward and score screens; maps for levels 2 and 3.
- **Programmer:** elites and keys, villages, shrines, events, hazards, lives and score.
- **Tester:** regression testing after each feature; looks for exploits in the rules.

Exit check: a full run from level 1 to the final boss is possible with no blocking bugs.

Stage 3. Playtests and balance (~20%)

- **PM / Game designer:** analyses playtest results; adjusts numbers and maps.
- **UI / Level designer:** fixes readability and layout issues found in playtests.
- **Programmer:** bug fixes; a debug panel and a fight simulator for balance checks.
- **Tester:** runs playtests with people outside the team; compares results with the design targets.

Exit check: a full run takes 35–45 minutes and no single route always wins.

Stage 4. Polish (~12%)

- **PM / Game designer:** chooses which nice-to-haves fit; final text pass.
- **UI / Level designer:** consistent art set, hit feedback, final UI pass.
- **Programmer:** implements chosen extras; performance and browser checks.
- **Tester:** full regression pass and a second playtest.

Exit check: feature freeze.

Stage 5. Release and defence (~8%)

- **PM / Game designer:** final design document, report and presentation.

- **UI / Level designer:** screenshots and visuals for the presentation.
- **Programmer:** release build and deployment; last fixes.
- **Tester:** final acceptance test against the must-have list.

Exit check: game deployed and the presentation ready.

7. Estimated hours

The whole project is estimated at about 316 hours across the team.

Stage	PM / GD	UI / LD	Programmer	Tester	Total
0. Kick-off	8	5	7	5	25
1. First level	14	20	32	10	76
2. Full game	18	22	34	14	88
3. Playtests	12	12	14	22	60
4. Polish	6	14	10	10	40
5. Release	8	5	6	8	27
Total, h	**66**	**78**	**103**	**69**	**316**

8. Working rules and risks

- A short weekly sync led by the PM; all tasks are tracked on one shared board.
- Every change is reviewed by another team member before it is merged.
- Game content is changed through data files, so balance can be tuned without code changes.
- Every stage ends with a playable build that the whole team tries.

Risk	Response
Balance takes longer than planned	Fight simulator and debug panel in stage 3; tune numbers before redesigning maps.
Scope creep from new ideas	New ideas go to the nice-to-have list; nothing new after the stage 4 feature freeze.
Inconsistent code or art style	One code style and one art direction agreed in stage 0; review before merge.